Lecture 10: Moore-Penrose Pseudoinverse

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Moore-penrose pseudoinverse
- Projectors onto measurement space and null space
- Projectors onto consistency space and inconsistency space

Penrose Equations

For a given linear operator $\mathcal{H}$, a **generalized inverse/pseudoinverse** $\mathcal{H}^\sharp$ satisfies the condition: $\mathcal{H}\mathcal{H}^\sharp\mathcal{H} = \mathcal{H}$.

The full set of **Penrose equations**:

$$\begin{aligned}
&\text{Penrose equation 1: } \mathcal{H}\mathcal{H}^\sharp\mathcal{H} = \mathcal{H} \\
&\text{Penrose equation 2: } \mathcal{H}^\sharp\mathcal{H}\mathcal{H}^\sharp = \mathcal{H}^\sharp \\
&\text{Penrose equation 3: } (\mathcal{H}\mathcal{H}^\sharp)^\dagger = \mathcal{H}\mathcal{H}^\sharp \\
&\text{Penrose equation 4: } (\mathcal{H}^\sharp\mathcal{H})^\dagger = \mathcal{H}^\sharp\mathcal{H}
\end{aligned}$$

A matrix that satisfies Penrose Equations 1 and 2 but not the other two is called a (1,2)-inverse.

Moore-Penrose Pseudoinverse

A **Moore-Penrose pseudoinverse**, denoted $\mathcal{H}^+$, is a specific generalized inverse that satisfies **all four Penrose equations**, i.e, a (1,2,3,4)-inverse.

The concept of pseudoinverse is not restricted to matrices and is applicable to any bounded, linear operator $\mathcal{H}$ mapping one separable Hilbert space to another.

Pseudoinverses and Singular-Value Decomposition (SVD)

Using the singular-value decomposition (SVD) of $\mathcal{H}$, the Moore-Penrose pseudoinverse $\mathcal{H}^+$ is given by:

$$\mathcal{H} = \sum_{k=1}^{R} \sqrt{\mu_k} \mathbf{v}_k \mathbf{u}_k^\dagger, \quad \mathcal{H}^+ = \sum_{k=1}^{R} \frac{1}{\sqrt{\mu_k}} \mathbf{u}_k \mathbf{v}_k^\dagger,$$

where

- $\mathbf{u}_k \in \mathbb{U}$: Right singular vectors of $\mathcal{H}$, orthonormal bases
- $\mathbf{v}_k \in \mathbb{V}$: Left singular vectors of $\mathcal{H}$, orthonormal bases
- $\sqrt{\mu_k} \geq 0$: Singular values of $\mathcal{H}$

💡 *The adjoint operator* $\mathcal{H}^\dagger = \sum_{k=1}^{R} \sqrt{\mu_k} \mathbf{u}_k \mathbf{v}_k^\dagger$.

Penrose Equation 1: $\mathcal{H}\mathcal{H}^\sharp\mathcal{H} = \mathcal{H}$

$$\mathcal{H}\mathcal{H}^+\mathcal{H} = \sum_{k=1}^{R} \sqrt{\mu_k} \mathbf{v}_k \mathbf{u}_k^\dagger \sum_{m=1}^{R} \frac{1}{\sqrt{\mu_m}} \mathbf{u}_m \mathbf{v}_m^\dagger \sum_{n=1}^{R} \sqrt{\mu_n} \mathbf{v}_n \mathbf{u}_n^\dagger$$

$$= \sum_{k=1}^{R}\sum_{m=1}^{R}\sum_{n=1}^{R} \frac{\sqrt{\mu_k}\sqrt{\mu_n}}{\sqrt{\mu_m}} \mathbf{v}_k \mathbf{u}_k^\dagger \mathbf{u}_m \mathbf{v}_m^\dagger \mathbf{v}_n \mathbf{u}_n^\dagger = \sum_{k=1}^{R}\sum_{m=1}^{R}\sum_{n=1}^{R} \frac{\sqrt{\mu_k}\sqrt{\mu_n}}{\sqrt{\mu_m}} \mathbf{v}_k \delta_{km} \delta_{mn} \mathbf{u}_n^\dagger$$

$$= \sum_{k=1}^{R} \frac{\sqrt{\mu_k}\sqrt{\mu_k}}{\sqrt{\mu_k}} \mathbf{v}_k \mathbf{u}_k^\dagger = \sqrt{\mu_k} \mathbf{v}_k \mathbf{u}_k^\dagger = \mathcal{H}$$

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\mathbf{p}_m^\dagger \mathbf{p}_n = \delta_{mn}$, i.e., $\mathbf{p}_m^\dagger \mathbf{p}_n = 1$, if $m = n$ and $\mathbf{p}_m^\dagger \mathbf{p}_n = 0$, if $m \neq n$

Penrose Equation 2: $\mathcal{H}^\sharp\mathcal{H}\mathcal{H}^\sharp = \mathcal{H}^\sharp$

Homework 3

Penrose Equation 3: $(\mathcal{H}\mathcal{H}^\sharp)^\dagger = \mathcal{H}\mathcal{H}^\sharp$

Homework 3

Penrose Equation 4: $(\mathcal{H}^\sharp\mathcal{H})^\dagger = \mathcal{H}^\sharp\mathcal{H}$

$$(\mathcal{H}\mathcal{H}^+)^\dagger = \left(\sum_{m=1}^{R}\frac{1}{\sqrt{\mu_m}}\mathbf{u}_m\mathbf{v}_m^\dagger\sum_{n=1}^{R}\sqrt{\mu_n}\mathbf{v}_n\mathbf{u}_n^\dagger\right)^\dagger$$

$$= \left(\sum_{m=1}^{R}\sum_{n=1}^{R}\frac{\sqrt{\mu_n}}{\sqrt{\mu_m}}\mathbf{u}_m\mathbf{v}_m^\dagger\mathbf{v}_n\mathbf{u}_n^\dagger\right)^\dagger = \left(\sum_{m=1}^{R}\sum_{n=1}^{R}\frac{\sqrt{\mu_n}}{\sqrt{\mu_m}}\mathbf{u}_m\delta_{mn}\mathbf{u}_n^\dagger\right)^\dagger$$

$$= \left(\sum_{m=1}^{R}\frac{\sqrt{\mu_m}}{\sqrt{\mu_m}}\mathbf{u}_m\mathbf{u}_m^\dagger\right)^\dagger = \left(\sum_{m=1}^{R}\mathbf{u}_m\mathbf{u}_m^\dagger\right)^\dagger = \sum_{m=1}^{R}\mathbf{u}_m\mathbf{u}_m^\dagger = \mathcal{H}\mathcal{H}^+$$

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\mathbf{p}_m^\dagger\mathbf{p}_n = \delta_{mn}$, i.e., $\mathbf{p}_m^\dagger\mathbf{p}_n = 1$, if $m = n$ and $\mathbf{p}_m^\dagger\mathbf{p}_n = 0$, if $m \neq n$

Properties of the Pseudoinverse

Limiting representation of the pseudoinverse will be used for iterative calculation of pseudoinverse solutions.

$$\mathcal{H}^+ = \lim_{\eta\to 0+}[\mathcal{H}^\dagger\mathcal{H} + \eta\mathcal{I}_\mathbb{U}]^{-1}\mathcal{H}^\dagger = \lim_{\eta\to 0+}\mathcal{H}^\dagger[\mathcal{H}\mathcal{H}^\dagger + \eta\mathcal{I}_\mathbb{V}]^{-1}$$

$$\mathcal{H}^\dagger\mathcal{H} + \eta\mathcal{I}_\mathbb{U} = \sum_{n=1}^{R}\mu_n\mathbf{u}_n\mathbf{u}_n^\dagger + \eta\sum_{n=1}^{N}\mathbf{u}_n\mathbf{u}_n^\dagger = \sum_{n=1}^{N}(\mu_n+\eta)\mathbf{u}_n\mathbf{u}_n^\dagger, \quad \mu_n = 0 \text{ for } n > R$$

$$[\mathcal{H}^\dagger\mathcal{H} + \eta\mathcal{I}_\mathbb{U}]^{-1} = \sum_{n=1}^{N}\frac{1}{\mu_n+\eta}\mathbf{u}_n\mathbf{u}_n^\dagger, \quad \mu_n \geq 0,\ \eta > 0$$

$$[\mathcal{H}^\dagger\mathcal{H}+\eta\mathcal{I}_\mathbb{U}]^{-1}\mathcal{H}^\dagger=\sum_{n=1}^{N}\frac{1}{\mu_n+\eta}\mathbf{u}_n\mathbf{u}_n^\dagger\sum_{k=1}^{R}\sqrt{\mu_k}\mathbf{u}_k\mathbf{v}_k^\dagger$$

$$=\sum_{n=1}^{N}\sum_{k=1}^{R}\frac{\sqrt{\mu_k}}{\mu_n+\eta}\mathbf{u}_n\mathbf{u}_n^\dagger\mathbf{u}_k\mathbf{v}_k^\dagger=\sum_{n=1}^{N}\sum_{k=1}^{R}\frac{\sqrt{\mu_k}}{\mu_n+\eta}\mathbf{u}_n\delta_{nk}\mathbf{v}_k^\dagger$$

$$=\sum_{n=1}^{N}\frac{\sqrt{\mu_n}}{\mu_n+\eta}\mathbf{u}_n\mathbf{v}_n^\dagger$$

💡 *The adjoint operator* $\mathcal{H}^\dagger=\sum_{k=1}^{R}\sqrt{\mu_k}\mathbf{u}_k\mathbf{v}_k^\dagger$.

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\mathbf{p}_m^\dagger\mathbf{p}_n=\delta_{mn}$, i.e., $\mathbf{p}_m^\dagger\mathbf{p}_n=1$, if $m=n$ and $\mathbf{p}_m^\dagger\mathbf{p}_n=0$, if $m\neq n$

$$\mathcal{H}^+=\lim_{\eta\to0+}[\mathcal{H}^\dagger\mathcal{H}+\eta\mathcal{I}_\mathbb{U}]^{-1}\mathcal{H}^\dagger=\lim_{\eta\to0+}\left[\sum_{n=1}^{N}\frac{\sqrt{\mu_n}}{\mu_n+\eta}\mathbf{u}_n\mathbf{v}_n^\dagger\right]$$

$$\lim_{\eta\to0+}\frac{\sqrt{\mu_n}}{\mu_n+\eta}=\begin{cases}\frac{1}{\sqrt{\mu_n}} & \text{if } \mu_n\neq0\\ 0 & \text{if } \mu_n=0\end{cases}$$

$$\mathcal{H}^+=\lim_{\eta\to0+}[\mathcal{H}^\dagger\mathcal{H}+\eta\mathcal{I}_\mathbb{U}]^{-1}\mathcal{H}^\dagger=\sum_{n=1}^{R}\frac{1}{\sqrt{\mu_n}}\mathbf{u}_n\mathbf{v}_n^\dagger=\mathcal{H}^+$$

⭐ *If* $R<N$, $\mu_{R+1}=\ldots=\mu_N=0$. *Therefore,* $\sum_{n=1}^{N}$ *becomes* $\sum_{n=1}^{R}$.

$$\mathcal{H}^+=\lim_{\eta\to0+}[\mathcal{H}^\dagger\mathcal{H}+\eta\mathcal{I}_\mathbb{U}]^{-1}\mathcal{H}^\dagger=\lim_{\eta\to0+}\mathcal{H}^\dagger[\mathcal{H}\mathcal{H}^\dagger+\eta\mathcal{I}_\mathbb{V}]^{-1}$$

$$\mathcal{H}\mathcal{H}^\dagger+\eta\mathcal{I}_\mathbb{V}=\sum_{n=1}^{R}\mu_n\mathbf{v}_n\mathbf{v}_n^\dagger+\eta\sum_{n=1}^{N}\mathbf{v}_n\mathbf{v}_n^\dagger=\sum_{n=1}^{N}(\mu_n+\eta)\mathbf{v}_n\mathbf{v}_n^\dagger,\quad \mu_n=0 \text{ for } n>R$$

$$[\mathcal{H}\mathcal{H}^\dagger+\eta\mathcal{I}_\mathbb{V}]^{-1}=\sum_{n=1}^{N}\frac{1}{\mu_n+\eta}\mathbf{v}_n\mathbf{v}_n^\dagger,\quad \mu_n\geq0,\ \eta>0$$

$$\mathcal{H}^\dagger[\mathcal{H}\mathcal{H}^\dagger+\eta\mathcal{I}_{\mathbb{V}}]^{-1}=\sum_{k=1}^{R}\sqrt{\mu_k}\mathbf{u}_k\mathbf{v}_k^\dagger\sum_{n=1}^{N}\frac{1}{\mu_n+\eta}\mathbf{v}_n\mathbf{v}_n^\dagger$$

$$=\sum_{n=1}^{N}\sum_{k=1}^{R}\frac{\sqrt{\mu_k}}{\mu_n+\eta}\mathbf{u}_k\mathbf{v}_k^\dagger\mathbf{v}_n\mathbf{v}_n^\dagger=\sum_{n=1}^{N}\sum_{k=1}^{R}\frac{\sqrt{\mu_k}}{\mu_n+\eta}\mathbf{u}_k\delta_{kn}\mathbf{v}_n^\dagger$$

$$=\sum_{n=1}^{N}\frac{\sqrt{\mu_n}}{\mu_n+\eta}\mathbf{u}_n\mathbf{v}_n^\dagger$$

💡 *The adjoint operator* $\mathcal{H}^\dagger=\sum_{k=1}^{R}\sqrt{\mu_k}\mathbf{u}_k\mathbf{v}_k^\dagger$.

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\mathbf{p}_m^\dagger\mathbf{p}_n=\delta_{mn}$, i.e., $\mathbf{p}_m^\dagger\mathbf{p}_n=1$, if $m=n$ and $\mathbf{p}_m^\dagger\mathbf{p}_n=0$, if $m\neq n$

$$\mathcal{H}^+=\lim_{\eta\to0+}\mathcal{H}^\dagger[\mathcal{H}\mathcal{H}^\dagger+\eta\mathcal{I}_{\mathbb{V}}]^{-1}\lim_{\eta\to0+}\left[\sum_{n=1}^{N}\frac{\sqrt{\mu_n}}{\mu_n+\eta}\mathbf{u}_n\mathbf{v}_n^\dagger\right]$$

$$\lim_{\eta\to0+}\frac{\sqrt{\mu_n}}{\mu_n+\eta}=\begin{cases}\frac{1}{\sqrt{\mu_n}} & \text{if } \mu_n\neq0\\ 0 & \text{if } \mu_n=0\end{cases}$$

$$\mathcal{H}^+=\lim_{\eta\to0+}\mathcal{H}^\dagger[\mathcal{H}\mathcal{H}^\dagger+\eta\mathcal{I}_{\mathbb{V}}]^{-1}=\sum_{n=1}^{R}\frac{1}{\sqrt{\mu_n}}\mathbf{u}_n\mathbf{v}_n^\dagger=\mathcal{H}^+$$

⭐ *If* $R<N$, $\mu_{R+1}=\ldots=\mu_N=0$. *Therefore,* $\sum_{n=1}^{N}$ *becomes* $\sum_{n=1}^{R}$.

Special Cases

Consider an $N\times N$ **Hermitian** matrix $\mathbf{H}$ ($\mathbf{v}_i=\mathbf{u}_i$ and $\lambda_j=\lambda_j^*$) with rank R. By its **spectral decomposition**, it can be expressed as:

$$\mathbf{H}=\sum_{j=1}^{R}\lambda_j\mathbf{u}_j\mathbf{u}_j^\dagger.$$

This expansion is a special case of SVD, with $\mu_j=\lambda_j^2$ ($\lambda_j\geq0$ and thus $\mathbf{H}\succeq0$).

Pseudoinverse of $\mathbf{H}$:

$$\mathbf{H}^+=\sum_{j=1}^{R}\frac{1}{\lambda_j}\mathbf{u}_j\mathbf{u}_j^\dagger$$

If $\mathbf{H}$ is **Hermitian** and **diagonal**, then:

$$\mathbf{H}=\operatorname{diag}(\lambda_1,\lambda_2,\ldots,\lambda_N).$$

Pseudoinverse of $\mathbf{H}$:

$$\mathbf{H}^{+} = \operatorname{diag}(\lambda_1^{+}, \lambda_2^{+}, \ldots, \lambda_N^{+}), \quad \lambda_j^{+} = \begin{cases} \frac{1}{\lambda_j} & \text{if } \lambda_j \neq 0 \\ 0 & \text{if } \lambda_j = 0 \end{cases}$$

Other Useful Properties

$$
\begin{aligned}
[\mathcal{H}^{+}]^{+} &= \mathcal{H} && && (1)\\
[\mathcal{H}^{\dagger}]^{+} &= [\mathcal{H}^{+}]^{\dagger} && && (2)\\
[\mathcal{H}^{+}\mathcal{H}]^{+} &= [\mathcal{H}]^{+}[\mathcal{H}^{+}]^{+} = \mathcal{H}^{+}\mathcal{H} && \text{(Using (1))} && (3)\\
[\mathcal{H}\mathcal{H}^{+}]^{+} &= [\mathcal{H}^{+}]^{+}[\mathcal{H}]^{+} = \mathcal{H}\mathcal{H}^{+} && \text{(Using (1))} && (4)\\
[\mathcal{H}^{\dagger}\mathcal{H}]^{+} &= [\mathcal{H}]^{+}[\mathcal{H}^{\dagger}]^{+} = \mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+} && && (5)\\
[\mathcal{H}\mathcal{H}^{\dagger}]^{+} &= [\mathcal{H}^{\dagger}]^{+}[\mathcal{H}]^{+} = [\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+} && && (6)\\
[\mathcal{H}^{\dagger}]^{+} &= [\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+}\mathcal{H} = \left[[\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+}\right]\mathcal{H} && && (1)\\
&= [\mathcal{H}\mathcal{H}^{\dagger}]^{+}\mathcal{H} && \text{(Using (6))} && (7)\\
\mathcal{H}^{+} &= \mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{\dagger} = \left[\mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+}\right]\mathcal{H}^{\dagger} && && (2)\\
&= [\mathcal{H}^{\dagger}\mathcal{H}]^{+}\mathcal{H}^{\dagger} && \text{(Using (5))} && (8.a)\\
&= \mathcal{H}^{\dagger}[\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+} = \mathcal{H}^{\dagger}\left[[\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+}\right] && && (3)\\
&= \mathcal{H}^{\dagger}[\mathcal{H}\mathcal{H}^{\dagger}]^{+} && \text{(Using (6))} && (8.b)\\
\mathcal{H}^{\dagger} &= \mathcal{H}^{+}\mathcal{H}\mathcal{H}^{\dagger} && && (9.a)\\
&= \mathcal{H}^{\dagger}\mathcal{H}\mathcal{H}^{+} && && (9.b)\\
[\mathcal{H}^{\dagger}\mathcal{H}]^{+} &= \mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+} = \mathcal{H}^{+}[\mathcal{H}\mathcal{H}^{\dagger}]^{+}\mathcal{H} && && (4)\\
&= \mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+}\mathcal{H} = \mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+} && \text{(Using (7))} && (10.a)\\
&= \mathcal{H}^{+}[\mathcal{H}^{\dagger}]^{+} = \mathcal{H}^{\dagger}[\mathcal{H}\mathcal{H}^{\dagger}]^{+}[\mathcal{H}^{\dagger}]^{+} && \text{(Using (8.b))} && (10.b)\\
[\mathcal{H}\mathcal{H}^{\dagger}]^{+} &= [\mathcal{H}^{\dagger}]^{+}\mathcal{H}^{+} && && (11)\\
\mathcal{H}^{+}\mathcal{H} &= [\mathcal{H}^{\dagger}\mathcal{H}]^{+}\mathcal{H}^{\dagger}\mathcal{H} = \mathcal{H}^{\dagger}\mathcal{H}[\mathcal{H}^{\dagger}\mathcal{H}]^{+} && \text{(Using (8.a))} && (12)\\
\mathcal{H}\mathcal{H}^{+} &= \mathcal{H}\mathcal{H}^{\dagger}[\mathcal{H}\mathcal{H}^{\dagger}]^{+} = [\mathcal{H}\mathcal{H}^{\dagger}]^{+}\mathcal{H}\mathcal{H}^{\dagger} && \text{(Using (8.b))} && (13)
\end{aligned}
$$

Pseudoinverse of a Product

Consider:

- An $M \times N$ matrix $\mathbf{A}$ of **full column rank** ($R = N$).
- An $N \times K$ matrix $\mathbf{B}$ of **full row rank** ($R = N$).

Under these conditions, the pseudoinverse of their product satifies:

$$[\mathbf{AB}]^{+} = \mathbf{B}^{+}\mathbf{A}^{+}.$$

⭐ *In general, the pseudoinverse of a product **does not equal** the product of the pseudoinverses. However, this holds when both matrices satisfy the given rank*

conditions above.

Pseudoinverses and Projection Operators

Consider the division of $\mathbb{U}$ into two **orthogonal** subspaces: the **measurement space** $\mathbb{U}_{meas}$ and the **null space** $\mathbb{U}_{null}$. Any arbitrary vector $\mathbf{f} \in \mathbb{U}$ can be expanded in terms of the **orthonormal** basis $\{\mathbf{u}_k\}$ as:

$$\mathbf{f} = \sum_{k=1}^{N} \alpha_k \mathbf{u}_k, \quad \text{where } \alpha_k = \mathbf{u}_k^\dagger \mathbf{f}$$

$$\mathcal{H}\mathbf{f} = \sum_{n=1}^{R} \sqrt{\mu_n}\mathbf{v}_n\mathbf{u}_n^\dagger \sum_{k=1}^{N} \alpha_k \mathbf{u}_k = \sum_{n=1}^{R}\sum_{k=1}^{N} \sqrt{\mu_n}\mathbf{v}_n\mathbf{u}_n^\dagger \alpha_k \mathbf{u}_k = \sum_{n=1}^{R}\sum_{k=1}^{N} \sqrt{\mu_n}\alpha_k \mathbf{v}_n \mathbf{u}_n^\dagger \mathbf{u}_k$$

$$= \sum_{n=1}^{R}\sum_{k=1}^{N} \sqrt{\mu_n}\alpha_k \mathbf{v}_n \delta_{nk} = \sum_{n=1}^{R} \sqrt{\mu_n}\alpha_n \mathbf{v}_n, \quad \mu_n = 0 \text{ for } n > R$$

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\mathbf{p}_m^\dagger \mathbf{p}_n = \delta_{mn}$, i.e., $\mathbf{p}_m^\dagger \mathbf{p}_n = 1$, if $m = n$ and $\mathbf{p}_m^\dagger \mathbf{p}_n = 0$, if $m \neq n$

$$\mathbf{f}_{null} = \sum_{k=R+1}^{N} \alpha_k \mathbf{u}_k, \quad \mathbf{f}_{meas} = \mathbf{f} - \mathbf{f}_{null} = \sum_{k=1}^{R} \alpha_k \mathbf{u}_k$$

When $n > R$ and $m \leq R$, $\mathbf{u}_m$ and $\mathbf{u}_n$ are orthogonal. It follows that: $\mathbf{f}_{meas} \perp \mathbf{f}_{null}$.

The operator $\mathcal{H}^+\mathcal{H}$ acts as a **projector operator** onto the **measurement space** $\mathbb{U}_{meas}$:

$$\mathcal{H}^+\mathcal{H} = \sum_{k=1}^{R} \frac{1}{\sqrt{\mu_k}}\mathbf{u}_k\mathbf{v}_k^\dagger \sum_{m=1}^{R} \sqrt{\mu_m}\mathbf{v}_m\mathbf{u}_m^\dagger = \sum_{k=1}^{R}\sum_{m=1}^{R} \frac{\sqrt{\mu_m}}{\sqrt{\mu_k}}\mathbf{u}_k\mathbf{v}_k^\dagger\mathbf{v}_m\mathbf{u}_m^\dagger$$

$$= \sum_{k=1}^{R}\sum_{m=1}^{R} \frac{\sqrt{\mu_m}}{\sqrt{\mu_k}}\mathbf{u}_k\delta_{km}\mathbf{u}_m^\dagger = \sum_{k=1}^{R} \mathbf{u}_k\mathbf{u}_k^\dagger = \mathcal{P}_{meas}$$

$$\mathcal{H}^+\mathcal{H}\mathbf{f} = \mathcal{P}_{meas}\mathbf{f} = \sum_{k=1}^{R} \mathbf{u}_k\mathbf{u}_k^\dagger \sum_{n=1}^{N} \alpha_n\mathbf{u}_n$$

$$= \sum_{k=1}^{R}\sum_{n=1}^{N} \alpha_n\mathbf{u}_k\mathbf{u}_k^\dagger\mathbf{u}_n = \sum_{k=1}^{R}\sum_{n=1}^{N} \alpha_n\mathbf{u}_k\delta_{kn} = \sum_{k=1}^{R} \alpha_k\mathbf{u}_k = \mathbf{f}_{meas}$$

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\mathbf{p}_m^\dagger \mathbf{p}_n = \delta_{mn}$, i.e., $\mathbf{p}_m^\dagger \mathbf{p}_n = 1$, if $m = n$ and $\mathbf{p}_m^\dagger \mathbf{p}_n = 0$, if $m \neq n$

$$\mathbf{f}_{null} = \mathbf{f} - \mathbf{f}_{meas} = (\mathcal{I}_{\mathbb{U}} - \mathcal{P}_{meas})\mathbf{f} = \mathcal{P}_{null}\mathbf{f}$$

$$\mathcal{P}_{null} = \mathcal{I}_{\mathbb{U}} - \mathcal{P}_{meas} = \mathcal{I}_{\mathbb{U}} - \mathcal{H}^{+}\mathcal{H}$$

$$\mathcal{P}_{meas} + \mathcal{P}_{null} = \sum_{k=1}^{R} \mathbf{u}_k \mathbf{u}_k^{\dagger} + \sum_{k=R+1}^{N} \mathbf{u}_k \mathbf{u}_k^{\dagger} = \sum_{k=1}^{N} \mathbf{u}_k \mathbf{u}_k^{\dagger} = \mathcal{I}_{\mathbb{U}}$$

$$\mathcal{H}\mathcal{P}_{null}\mathbf{f} = \mathcal{H}[\mathcal{I}_{\mathbb{U}} - \mathcal{H}^{+}\mathcal{H}]\mathbf{f} = [\mathcal{H} - \mathcal{H}\mathcal{H}^{+}\mathcal{H}]\mathbf{f} = [\mathcal{H} - \mathcal{H}]\mathbf{f} = \mathbf{0}$$

💡 Penrose equation 1: $\mathcal{H}\mathcal{H}^{\sharp}\mathcal{H} = \mathcal{H}$

Consider the division of $\mathbb{V}$ into two **orthogonal** subspaces: the **consistency space** $\mathbb{V}_{con}$ and the **inconsistency space** $\mathbb{V}_{incon}$. A similar decomposition holds for an arbitrary vector $\mathbf{g} \in \mathbb{V}$, expanded in terms of the **orthonormal** basis $\{\mathbf{v}_k\}$:

$$\mathbf{g} = \sum_{k=1}^{M} \beta_k \mathbf{v}_k = \mathbf{g}_{con} + \mathbf{g}_{incon}, \quad \text{where } \beta_k = \mathbf{v}_k^{\dagger}\mathbf{g}$$

The operator $\mathcal{H}\mathcal{H}^{+}$ acts as a **projector operator** onto the **consistency space** $\mathbb{V}_{con}$:

$$\mathcal{H}\mathcal{H}^{+} = \sum_{k=1}^{R} \sqrt{\mu_k}\mathbf{v}_k\mathbf{u}_k^{\dagger} \sum_{m=1}^{R} \frac{1}{\sqrt{\mu_m}}\mathbf{u}_m\mathbf{v}_m^{\dagger} = \sum_{k=1}^{R}\sum_{m=1}^{R} \frac{\sqrt{\mu_k}}{\sqrt{\mu_m}}\mathbf{v}_k\mathbf{u}_k^{\dagger}\mathbf{u}_m\mathbf{v}_m^{\dagger}$$

$$= \sum_{k=1}^{R}\sum_{m=1}^{R} \frac{\sqrt{\mu_k}}{\sqrt{\mu_m}}\mathbf{v}_k\delta_{km}\mathbf{v}_m^{\dagger} = \sum_{k=1}^{R} \mathbf{v}_k\mathbf{v}_k^{\dagger} = \mathcal{P}_{con}$$

$$\mathcal{H}\mathcal{H}^{+}\mathbf{g} = \mathcal{P}_{con}\mathbf{g} = \sum_{k=1}^{R} \mathbf{v}_k\mathbf{v}_k^{\dagger} \sum_{m=1}^{M} \beta_m\mathbf{v}_m$$

$$= \sum_{k=1}^{R}\sum_{m=1}^{M} \beta_m\mathbf{v}_k\mathbf{v}_k^{\dagger}\mathbf{v}_m = \sum_{k=1}^{R}\sum_{m=1}^{M} \beta_m\mathbf{v}_k\delta_{kn} = \sum_{k=1}^{R} \beta_k\mathbf{v}_k = \mathbf{g}_{con}$$

$$\mathcal{P}_{con} = \mathcal{H}\mathcal{H}^{+}, \quad \mathcal{P}_{incon} = \mathcal{I}_{\mathbb{V}} - \mathcal{H}\mathcal{H}^{+}$$

$$\mathcal{P}_{con} + \mathcal{P}_{incon} = \sum_{m=1}^{R} \mathbf{v}_m\mathbf{v}_m^{\dagger} + \sum_{k=R+1}^{M} \mathbf{v}_k\mathbf{v}_k^{\dagger} = \sum_{m=1}^{M} \mathbf{v}_m\mathbf{v}_m^{\dagger} = \mathcal{I}_{\mathbb{V}}$$